

Contents

Statistical Power for Detecting Biomarker-Treatment Interactions in N-of-1 Trials: A Monte Carlo Simulation Study	1
Abstract	2
1. Introduction	2
1.1 Background	2
1.2 Research Questions	2
1.3 Theoretical Foundation	2
2. Methods	2
2.1 Trial Designs Evaluated	2
2.2 Three-Factor Response Model	3
2.3 Biomarker Moderation	3
2.4 Carryover Modeling	3
2.5 Two-Scenario Comparison	4
2.6 Simulation Parameters	5
2.7 Statistical Analysis	5
3. Implementation	6
3.1 Software Architecture	6
3.2 Key Functions	6
3.3 Reproducibility Infrastructure	6
3.4 Computational Requirements	6
4. Results	6
4.1 Primary Outputs	6
4.2 Key Findings	7
4.3 Design Comparison	7
5. Project Structure	7
6. Key Findings from Hendrickson et al. (2020)	7
6.1 Design Comparison Results	7
6.2 Critical Carryover Finding	7
6.3 Practical Design Recommendation	8
7. Extensions Beyond Hendrickson	8
8. Usage	8
Quick Start	8
Vignettes	8
9. Dependencies	9
10. Citation	9
11. License	9
12. Research Plan: Prazosin-PTSD Trial Analysis	9
12.1 Goal 1: Did Prazosin Improve PTSD Symptoms?	9
12.2 Goal 2: Characterize Carryover in Actual Data	11
12.3 Goal 3: Enhance Simulation Modularity for Carryover Analysis	14
12.4 Integration Timeline and Dependencies	16
12.5 Prazosin-Specific Considerations	17
12.6 Summary: Three-Goal Integration	18
Appendix A: Covariance Matrix Construction	18
Appendix B: Carryover Decay Models	18

Statistical Power for Detecting Biomarker-Treatment Interactions in N-of-1 Trials: A Monte Carlo Simulation Study

pmsimstats2025 Project Summary

Abstract

This document summarizes the pmsimstats2025 simulation framework, which evaluates statistical power for detecting biomarker-treatment interactions across multiple N-of-1 trial designs. Building on Hendrickson et al. (2020), the framework extends the methodology to explicitly model carryover effects and quantify the statistical cost of ignoring carryover in analysis. Through Monte Carlo simulation comparing five trial designs (Hybrid, OL+BDC, Open-Label, Crossover, and Parallel), the project demonstrates that explicit carryover modeling preserves statistical power across varying carryover conditions, while omitting carryover terms leads to progressive power degradation.

1. Introduction

1.1 Background

Precision medicine seeks to optimize treatment selection based on individual patient characteristics. Biomarker-guided treatment represents a promising approach, but detecting biomarker-treatment interactions requires adequate statistical power. N-of-1 trials offer a rigorous framework for evaluating individual treatment effects, yet design choices significantly influence the ability to detect moderating effects.

1.2 Research Questions

1. How does statistical power to detect biomarker-treatment interactions compare across hybrid and traditional trial designs?
2. What is the impact of carryover effects on power estimation?
3. What is the statistical cost of ignoring carryover in the analysis model?

1.3 Theoretical Foundation

This work builds directly on Hendrickson et al. (2020), “Optimizing Aggregated N-Of-1 Trial Designs for Predictive Biomarker Validation: Statistical Methods and Theoretical Findings” (Frontiers in Digital Health, 2:13), implementing their three-factor response model (BR + ER + TR) and extending it with explicit carryover modeling in the analysis phase.

The Hendrickson framework addresses a specific question: whether a baseline biomarker (standing systolic blood pressure) predicts individual response to prazosin for PTSD. The simulation approach generates correlated multivariate normal data representing biological, expectancy, and time-variant response components, then tests for significant biomarker $\times$ treatment interactions.

2. Methods

2.1 Trial Designs Evaluated

The framework compares five distinct trial designs:

Design	Measurement Schedule (weeks)	Points	Structure
Hybrid	4, 8, 9, 10, 11, 12, 16, 20	8	Dense cluster at phase transition
OL+BDC	4, 8, 12, 16, 17, 18, 19, 20	8	Dense at discontinuation
Open-Label	2, 8, 14, 20	4	Evenly spaced (6-week intervals)
Crossover	2, 8, 14, 20	4	Traditional AB/BA sequence
Parallel	2, 8, 14, 20	4	Between-subjects comparison

2.2 Three-Factor Response Model

The simulation generates participant responses as the sum of three components:

$$\text{Response} = \text{Baseline} + \text{BR} + \text{ER} + \text{TR}$$

Where:

- **Biological Response (BR):** Accumulates during active treatment, moderated by biomarker level. Includes carryover with exponential decay.
- **Expectancy Response (ER):** Based on treatment visibility, scaled by participant knowledge of treatment status (1.0 for open-label when participant knows they are on active drug, 0.5 for blinded conditions where participant believes there is 50% chance of active drug).
- **Time-variant Response (TR):** Non-treatment-specific improvement over trial duration, including regression to mean and effects of study participation (contact with staff, symptom monitoring). Per Hendrickson's "tabula rasa" assumption, TR and ER are assumed to have equal magnitude and variance in the absence of empirical data to distinguish them.

2.3 Biomarker Moderation

The biomarker-treatment interaction is modeled as:

$$\text{BR_rate} \times (1 + \beta_{\text{mod}} \times \text{biomarker_centered})$$

Where β_{mod} represents the moderation strength, varied systematically from 0 to 0.65 across simulation conditions.

2.4 Carryover Modeling

Carryover effects follow exponential decay, applied only to the biological response (BR) component. The framework supports a unified parameterization that accommodates both half-life and proportion-based specifications.

2.4.1 Exponential Decay Model The primary carryover model uses time-since-discontinuation (tsd):

$$\text{carryover_retention} = (1/2)^{(\text{tsd} / \text{t_half})}$$

Where: - tsd: weeks elapsed since participant discontinued active treatment - t_half: half-life of carryover effect in weeks

2.4.2 Unified Parameterization The framework supports two equivalent parameterization modes:

Mode	Parameter	Formula	Use Case
half-life	$t_{1/2}$ in weeks	$(1/2)^{(tsd / t_{half})}$	Pharmacokinetic modeling
proportion	p (0-1)	Fixed retention proportion	Simple sensitivity analysis

Conversion between modes (at reference time t): - Proportion $\rightarrow$ Half-life: $t_{half} = -t / \log_2(p)$ - Half-life $\rightarrow$ Proportion: $p = (1/2)^{(t / t_{half})}$

Example: 50% retention at 1 week $\leftrightarrow t_{1/2} = 1$ week

2.4.3 Implementation Details For each participant, the simulation:

1. Tracks when treatment is discontinued (discontinuation_week)
2. Computes tsd at each subsequent timepoint
3. Applies exponential decay to accumulated BR at discontinuation:

$$BR_{off_drug} = BR_{at_discontinuation} \times (1/2)^{(tsd / t_{half})}$$
4. Generates continuous carryover_effect covariate for analysis model

2.4.4 Scale Considerations Critical Note: Hendrickson et al. (2020) used very short half-lives ($t_{1/2} = 0, 0.1, 0.2$ weeks, i.e., 0.7-1.4 days), reflecting rapid pharmacokinetic washout. The current implementation explores longer half-lives (0, 0.5, 1.0, 2.0 weeks) to capture potential neurobiological adaptation effects that persist beyond plasma drug clearance.

Scale	$t_{1/2}$ Range	Interpretation
Hendrickson	0-0.2 weeks	Plasma drug clearance
pmsimstats	0-2.0 weeks	Clinical effect persistence
Prazosin PK	~ 0.02 weeks	Plasma half-life (2-3 hours)
PTSD effect	Unknown	To be estimated from trial data (Section 12)

Parameters: - scale_factor: 1.0 (biological response only, per Hendrickson) - t_{half} (half-life): 0, 0.5, 1.0, or 2.0 weeks

2.5 Two-Scenario Comparison

The core methodological contribution compares analysis approaches. Hendrickson et al. (2020) modeled carryover in data generation but did not include it in the analysis model—they observed that even short carryover caused “precipitous decline in power” for N-of-1 designs. This framework extends their work by explicitly testing whether including carryover in the analysis model recovers lost power.

Scenario 1 (WITH Carryover Modeling):

```
lmer(response ~ treatment * biomarker + week + carryover_effect +
      (1 | participant_id))
```

Scenario 2 (WITHOUT Carryover Modeling) — Hendrickson’s original approach:

```
lmer(response ~ treatment * biomarker + week +
      (1 | participant_id))
```

Note on Design-Specific Models: Following Hendrickson, open-label-only designs use a time $\times$ biomarker interaction model rather than treatment $\times$ biomarker, since all participants experience the same treatment condition:

```
lmer(response ~ biomarker + week + biomarker:week + (1 | participant_id))
```

2.6 Simulation Parameters

Fixed Parameters:

Parameter	Value	Description
n_participants	70	Sample size per simulation
n_iterations	100	Monte Carlo replications
BR_rate	0.5	Biological response rate (pts/week)
ER_rate	0.2	Expectancy rate (pts/week)
TR_rate	0.1	Time-variant rate (pts/week)
baseline_mean	10.0	Mean baseline score
within_subject_sd	1.8	Within-subject variability
between_subject_sd	2.0	Between-subject variability

Correlation Structure (Hendrickson values):

Parameter	Value	Description
c.br	0.75	Biological response autocorrelation
c.er	0.75	Expectancy response autocorrelation
c.tr	0.75	Time-variant response autocorrelation
c.cf1t	0.12	Same-time cross-correlation
c.cfct	0.05	Different-time cross-correlation
c.bm_baseline	0.25	Biomarker-baseline correlation

Parameter Grid (Two-Scenario Design):

```
expand_grid(
  design = c("hybrid", "ol_bdc", "open_label", "crossover", "parallel"),
  biomarker_moderation = c(0, 0.25, 0.35, 0.45, 0.55, 0.65),
  biomarker_correlation = 0.3,
  carryover_half-life = c(0, 0.5, 1.0, 2.0),
  model_carryover = c(TRUE, FALSE)
)
```

The `model_carryover` parameter enables the two-scenario comparison: - TRUE: Include carryover_effect term in analysis model - FALSE: Omit carryover term (replicates Hendrickson approach)

When `carryover_half-life = 0`, the `model_carryover` parameter is redundant (no carryover to model), so these duplicate rows are filtered from the grid.

2.7 Statistical Analysis

Power is computed as the proportion of simulations yielding a significant treatment $\times$ biomarker interaction ($p < 0.05$) from linear mixed-effects models with random intercepts for participants and CAR(1) correlation structure for repeated measures.

3. Implementation

3.1 Software Architecture

The framework is implemented as an R package (pmsimstats2025) with:

- **Core Functions** (pm_functions.R, 1,301 lines): Data generation, covariance matrix construction, carryover calculation, and response modeling
- **Simulation Scripts**: Separate scripts for clustered designs (simulation_clustered.R) and evenly-spaced designs (simulation_evenly_spaced.R)
- **Visualization**: Publication-ready power heatmaps in Hendrickson style

3.2 Key Functions

Function	Purpose
build_sigma_matrix()	Construct positive-definite covariance matrix
generate_data()	Generate multivariate normal trial data
calculate_carryover()	Model carryover decay (exponential, linear, Weibull)
calculate_bio_response_with_interaction()	Compute biomarker-moderated response
validate_parameter_grid()	Pre-simulation parameter validation

3.3 Reproducibility Infrastructure

- **Docker**: Containerized environment via zzzcollab framework
- **renv**: Locked dependencies (252 KB lock file)
- **Makefile**: 229 lines of automation targets
- **Testing**: Unit tests (testthat) and integration tests

3.4 Computational Requirements

Configuration	Runtime
20 iterations	5-15 minutes
100 iterations	~25 minutes
1,000 iterations	~4 hours

(M1 Mac, 16GB RAM baseline)

4. Results

4.1 Primary Outputs

The framework generates:

1. **Power Heatmaps**: Two-dimensional surfaces showing statistical power across biomarker correlation and carryover half-life parameters
2. **Coefficient Summaries**: Mixed model parameter estimates across conditions
3. **Publication Figures**: Hendrickson-style two-panel comparisons (WITH vs. WITHOUT carryover modeling)

4.2 Key Findings

The two-scenario comparison demonstrates:

- **With Carryover Modeling:** Statistical power remains stable across carryover conditions, as the confounding effect is explicitly controlled
- **Without Carryover Modeling:** Progressive power degradation occurs as carryover half-life increases, due to unmodeled confounding

4.3 Design Comparison

Preliminary results indicate that hybrid designs with dense measurement clusters at phase transitions provide enhanced sensitivity for detecting biomarker- treatment interactions compared to traditional evenly-spaced designs.

5. Project Structure

```
pmsimstats2025/
├── analysis/
│   ├── scripts/           # Simulation and visualization (47 files)
│   ├── output/            # Results and figures
│   ├── data/              # Raw and derived data
│   └── report/            # Manuscript template
├── docs/                  # Documentation (20 files)
├── vignettes/             # Tutorials (4 vignettes)
├── R/                     # Package functions
├── tests/                 # Test suite
├── renv/                  # Dependency management
├── Dockerfile             # Container specification
├── Makefile               # Build automation
└── README.md              # Project overview
```

6. Key Findings from Hendrickson et al. (2020)

The foundational paper established several critical findings that inform this framework:

6.1 Design Comparison Results

Design	Power (no carryover)	Power ($t_{1/2}=0.1$ wk)	Notes
Open-Label	Low	Low	No within-subject contrast
OL+BDC	Moderate	Sharply reduced	Vulnerable to carryover
Crossover	Highest	Moderately reduced	Most robust to carryover
N-of-1	High	Sharply reduced	Best balance of power + practicality

6.2 Critical Carryover Finding

“The presence of even a short (0.1 weeks) carryover component resulted in a precipitous decline in power in both the N-of-1 and, to a lesser but still very signifi-

cant degree, the open label + blinded discontinuation design.” — Hendrickson et al. (2020), p. 7

This finding motivates the central question of pmsimstats2025: can explicit carryover modeling in the analysis phase recover the lost power?

6.3 Practical Design Recommendation

Hendrickson recommended the N-of-1 design (8 weeks open-label → 4 weeks blinded discontinuation → 8 weeks crossover) despite slightly lower power than traditional crossover because it allows all participants to start on active treatment—critical for recruiting symptomatic patients who would otherwise decline randomization to placebo.

7. Extensions Beyond Hendrickson

Feature	Hendrickson (2020)	This Framework
Three-factor model (BR+ER+TR)	Core	Implemented
BR-only carryover in data gen	Core	Implemented
Expectancy scaling (1.0/0.5)	Core	Implemented
Tabula rasa assumption (TR=ER)	Core	Implemented
Carryover half-life range	0, 0.1, 0.2 weeks	0, 0.5, 1.0, 2.0 weeks
Carryover in analysis model	Not included	Two-scenario comparison
Biomarker moderation grid	Fixed at observed values	Systematic variation
Parameter validation	Not described	Pre-simulation PD checks
Design-specific models	Implicit	Explicit documentation

Key Methodological Extension: Hendrickson observed that carryover caused “precipitous decline in power” but did not test whether including carryover in the analysis model could recover this power. This framework directly addresses that gap through the two-scenario comparison.

8. Usage

Quick Start

```
# Enter Docker environment
make r
```

```
# Run clustered design simulations
Rscript analysis/scripts/simulation_clustered.R
```

```
# Generate visualizations
Rscript analysis/scripts/visualize_hendrickson_style.R
```

Vignettes

1. 01_generate_simulation_data.Rmd - Data generation workflow
2. 02_analyze_and_visualize.Rmd - Analysis and visualization
3. 03_apply_to_clinical_data.Rmd - Clinical application

4. quickstart.Rmd - Rapid orientation

9. Dependencies

Core Packages:

- Statistical modeling: lme4, lmerTest, nlme
 - Data manipulation: dplyr, tidyverse, data.table
 - Visualization: ggplot2, patchwork, viridis
 - Matrix operations: MASS, corpcor
 - Parallel processing: foreach, doParallel, future, furrr
-

10. Citation

```
@software{pmsimstats2025,  
  title = {pmsimstats2025: Monte Carlo Simulation for N-of-1 Trial Design},  
  year = {2025},  
  url = {https://github.com/[repository]},  
  note = {R package version 1.0.0}  
}
```

Key Reference:

Hendrickson RC, Thomas RG, Schork NJ, Raskind MA (2020). Optimizing Aggregated N-Of-1 Trial Designs for Predictive Biomarker Validation: Statistical Methods and Theoretical Findings. *Frontiers in Digital Health*, 2:13. doi: 10.3389/fdgth.2020.00013

Related Clinical Trial: NCT03539614 (VA-funded N-of-1 trial of prazosin for PTSD, currently enrolling)

11. License

GPL-3

12. Research Plan: Prazosin-PTSD Trial Analysis

This section outlines the methodological plan for applying the pmsimstats2025 framework to analyze actual prazosin trial data for PTSD treatment, with three primary goals: (1) evaluating treatment efficacy, (2) characterizing carryover effects, and (3) enhancing simulation modularity.

12.1 Goal 1: Did Prazosin Improve PTSD Symptoms?

12.1.1 Analysis Challenge The hybrid N-of-1 design presents unique analytical challenges compared to traditional parallel-group RCTs:

- Participants serve as their own controls across treatment phases
- The 4-path randomization structure creates heterogeneous treatment sequences
- Carryover effects may confound treatment phase comparisons
- Biomarker moderation requires interaction testing, not simple group means

12.1.2 Proposed Analysis Strategy Primary Analysis Model:

```
lmer(ptsd_score ~ treatment * biomarker_centered + week + carryover_effect +  
      (1 | participant_id),  
      data = prazosin_data)
```

This model estimates:

- treatment: Main effect of prazosin vs. placebo/off-drug
- biomarker_centered: Biomarker association with outcome
- treatment:biomarker_centered: Differential treatment effect by biomarker (primary hypothesis)
- week: Linear time trend (natural symptom trajectory)
- carryover_effect: Residual drug effect during washout periods
- (1 | participant_id): Between-subject heterogeneity

Interpretation Framework:

Coefficient	Interpretation if Significant ($p < 0.05$)
treatment	Average prazosin effect across biomarker range
treatment:biomarker	Biomarker moderates treatment response
carryover_effect	Residual effects persist after discontinuation

12.1.3 Secondary Analyses

1. **Path-Specific Effects:** Stratify by randomization path to assess treatment sequence effects

```
lmer(ptsd_score ~ treatment * biomarker * path + week + carryover +  
      (1 | participant_id))
```

2. **Sensitivity to Carryover Specification:** Compare results with and without carryover term to quantify confounding risk (mirrors simulation two-scenario comparison)
3. **Individual Treatment Effects:** Extract participant-level BLUPs (Best Linear Unbiased Predictors) for personalized effect estimates

```
ranef(model)$participant_id
```

4. **Responder Analysis:** Classify participants by biomarker-predicted response magnitude

12.1.4 Required Data Elements

Variable	Description	Source
participant_id	Unique identifier	Trial database
path	Randomization path (1-4)	Randomization
week	Assessment timepoint	Protocol
treatment	On prazosin (1) vs. off (0)	Protocol/path
ptsd_score	PTSD symptom measure (e.g., PCL-5, CAPS-5)	Assessment
biomarker	Pre-specified moderator variable	Baseline
tsd	Time since discontinuation (weeks)	Derived
carryover_effect	Computed decay function	Derived

12.1.5 Implementation Steps

1. Import and validate clinical dataset structure
 2. Compute treatment status indicators from path and week
 3. Calculate time-since-discontinuation (tsd) for each observation
 4. Apply carryover decay function (requires half-life specification)
 5. Center biomarker variable at sample mean
 6. Fit primary mixed-effects model
 7. Extract coefficients, confidence intervals, and p-values
 8. Conduct sensitivity analyses
 9. Generate effect plots (treatment effect as function of biomarker)
-

12.2 Goal 2: Characterize Carryover in Actual Data

12.2.1 The Carryover Quantification Problem Carryover effects in prazosin trials reflect residual drug activity after discontinuation. Characterizing carryover requires addressing:

1. **Temporal dynamics:** How quickly does the effect decay?
2. **Magnitude:** What proportion of on-drug effect persists?
3. **Individual variation:** Does carryover differ across participants?
4. **Outcome specificity:** Do different PTSD domains show different carryover?

12.2.2 Proposed Carryover Quantification Structure A. Pharmacokinetic Anchoring

Prazosin pharmacokinetics provide initial parameter bounds:

Parameter	Value	Implication
Plasma half-life	2-3 hours	Drug cleared within 24 hours
CNS effect duration	8-12 hours	Direct pharmacological effect
Symptom response lag	Days to weeks	Clinical effect may persist longer

The disconnect between plasma kinetics (hours) and symptom response (weeks) suggests the relevant carryover operates at the neurobiological adaptation level, not plasma drug concentration.

B. Empirical Carryover Estimation

Two complementary approaches:

Approach 1: Model-Based Estimation

Treat half-life as a parameter to estimate from data:

```
# Grid search over candidate half-lives
half_lives <- c(0.5, 1, 2, 3, 4, 6, 8) # weeks

results <- map_dfr(half_lives, function(t_half) {
  data <- prazosin_data |>
    mutate(carryover = ifelse(tsd > 0,
                              (1/2)^(tsd / t_half),
                              0))

  model <- lmer(ptsd_score ~ treatment * biomarker + week + carryover +
    (1 | participant_id), data = data)
```

```

tibble(
  t_half = t_half,
  AIC = AIC(model),
  BIC = BIC(model),
  carryover_coef = fixef(model)["carryover"],
  carryover_se = sqrt(vcov(model)["carryover", "carryover"])
)
})

```

Select half-life minimizing AIC/BIC or maximizing carryover coefficient precision.

Approach 2: Non-Parametric Visualization

Examine raw response patterns during washout periods:

```

prazosin_data |>
  filter(tsd > 0) |> # Post-discontinuation observations only
  ggplot(aes(x = tsd, y = ptsd_score, group = participant_id)) +
  geom_line(alpha = 0.3) +
  geom_smooth(aes(group = NULL), method = "loess") +
  labs(x = "Weeks Since Discontinuation",
       y = "PTSD Score",
       title = "Empirical Carryover Decay Pattern")

```

C. Multi-Model Comparison

Compare decay functional forms:

Model	Formula	Parameters
Exponential	$(1/2)^{(tsd / t_half)}$	t_half
Linear	$\max(0, 1 - tsd / t_total)$	t_total
Weibull	$\exp(-(tsd / \lambda)^k)$	λ, k
Step	$\text{ifelse}(tsd < \text{threshold}, 1, 0)$	threshold

Use likelihood ratio tests or AIC comparison to select best-fitting model.

12.2.3 Identification of Optimal Response Measure Sequence A. The Sequence Selection Problem

PTSD assessments may include multiple instruments or subscales:

- PCL-5 total score
- PCL-5 cluster subscores (intrusion, avoidance, cognition/mood, arousal)
- CAPS-5 clinician rating
- Secondary measures (sleep quality, nightmare frequency)

Different measures may show different:

- Sensitivity to treatment
- Carryover persistence
- Biomarker moderation patterns

B. Proposed Selection Strategy

Step 1: Univariate Screening

For each candidate outcome measure:

```

screen_outcome <- function(outcome_var, data) {
  formula <- as.formula(paste(outcome_var,
    "~ treatment * biomarker + week + (1 | participant_id)"))
  model <- lmer(formula, data = data)

  tibble(
    outcome = outcome_var,
    treatment_effect = fixef(model)["treatment"],
    interaction_effect = fixef(model)["treatment:biomarker"],
    interaction_p = summary(model)$coefficients["treatment:biomarker", "Pr(>|t|)"],
    icc = performance::icc(model)$ICC_adjusted,
    r2_marginal = performance::r2(model)$R2_marginal
  )
}

```

Step 2: Carryover Sensitivity Comparison

For each outcome, compare carryover model fit:

```

compare_carryover_sensitivity <- function(outcome_var, data) {
  # Model without carryover
  m0 <- lmer(paste(outcome_var, "~ treatment * biomarker + week +
    (1 | participant_id)"), data = data)

  # Model with carryover
  m1 <- lmer(paste(outcome_var, "~ treatment * biomarker + week + carryover +
    (1 | participant_id)"), data = data)

  tibble(
    outcome = outcome_var,
    delta_AIC = AIC(m0) - AIC(m1), # Positive = carryover helps
    carryover_coef = fixef(m1)["carryover"],
    lr_test_p = anova(m0, m1)$`Pr(>Chisq)`[2]
  )
}

```

Step 3: Composite Ranking

Rank outcomes by weighted criteria:

Criterion	Weight	Rationale
Interaction effect size	0.30	Primary hypothesis sensitivity
Carryover model improvement	0.25	Carryover characterization utility
ICC (reliability)	0.20	Measurement precision
Variance explained (R^2)	0.15	Signal-to-noise ratio
Clinical interpretability	0.10	Practical relevance

C. Recommended Sequence

Based on typical PTSD trial structure, prioritize:

1. **Primary:** PCL-5 total score (validated, comparable across studies)
2. **Secondary:** CAPS-5 clinician rating (gold standard, if available)
3. **Exploratory:** PCL-5 arousal cluster (most likely to show prazosin-specific effects given mechanism of action)
4. **Mechanistic:** Nightmare frequency (direct target of prazosin)

12.2.4 Deliverables for Goal 2

1. Estimated carryover half-life with confidence interval
 2. Comparison of decay model functional forms
 3. Ranked list of outcome measures by carryover sensitivity
 4. Visualization of empirical carryover decay patterns
 5. Participant-level carryover heterogeneity assessment
-

12.3 Goal 3: Enhance Simulation Modularity for Carryover Analysis

12.3.1 Current Implementation Status The existing `calculate_carryover()` function in `pm_functions.R` supports three decay models (exponential, linear, Weibull) but has limited modularity:

Current Limitations:

- Carryover parameters embedded in simulation scripts, not configuration files
- Single half-life value applied uniformly (no individual variation)
- No mechanism to import empirically-estimated carryover from real data
- Limited integration with sensitivity analysis workflows

12.3.2 Proposed Modular Architecture A. Carryover Configuration Object

Create a dedicated configuration structure:

```
carryover_config <- list(  
  # Model specification  
  model = "exponential",           # "exponential", "linear", "weibull", "step"  
  
  # Core parameters  
  half_life = 1.0,                 # weeks (can be vector for sensitivity)  
  half_life_sd = 0.3,              # between-subject SD (optional)  
  
  # Weibull-specific  
  shape = 1.0,                     # shape parameter k  
  
  # Component application  
  apply_to = c("BR"),              # which response components  
  scale_factors = c(BR = 1.0, ER = 0.0, TR = 0.0),  
  
  # Estimation mode  
  source = "specified",            # "specified", "estimated", "imported"  
  estimated_from = NULL,           # path to estimation results  
  
  # Sensitivity grid  
  sensitivity_grid = c(0, 0.5, 1.0, 2.0) # half-lives to test  
)
```

B. Modular Function Interface

Refactor `calculate_carryover()` to accept configuration object:

```
calculate_carryover <- function(tsd, config = carryover_config) {
```

```

# Validate inputs

stopifnot(config$model %in% c("exponential", "linear", "weibull", "step"))
stopifnot(tsd >= 0)

# Individual-level half-life (if heterogeneity specified)
t_half <- if (!is.null(config$half_life_sd)) {
  rnorm(1, config$half_life, config$half_life_sd) |> max(0.01)
} else {
  config$half_life
}

# Calculate decay
decay <- switch(config$model,
  exponential = (1/2)^(tsd / t_half),
  linear = pmax(0, 1 - tsd / t_half),
  weibull = exp(-(tsd / t_half)^config$shape),
  step = as.numeric(tsd < t_half)
)

decay
}

```

C. Integration with Empirical Estimates

Create import pathway for real-data carryover estimates:

```

import_carryover_estimate <- function(estimation_results_path) {
  # Load estimation results from Goal 2 analysis
  est <- readRDS(estimation_results_path)

  carryover_config <- list(
    model = est$best_model,
    half_life = est$half_life_estimate,
    half_life_sd = est$half_life_se,
    source = "estimated",
    estimated_from = estimation_results_path
  )

  carryover_config
}

```

D. Sensitivity Analysis Wrapper

Systematic sensitivity analysis across carryover specifications:

```

run_carryover_sensitivity <- function(base_config, half_life_grid,
                                     n_iterations = 100) {

  results <- map_dfr(half_life_grid, function(t_half) {
    config <- modifyList(base_config, list(half_life = t_half))

    # Run simulation with this carryover specification
    sim_result <- run_simulation(
      carryover_config = config,
      n_iterations = n_iterations
    )
  })
}

```

```

    sim_result |>
      mutate(carryover_half_life = t_half)
  })

  results
}

```

12.3.3 Implementation Roadmap Phase 1: Configuration Externalization

1. Define carryover_config schema in R/carryover_config.R
2. Create YAML configuration file support (config/carryover.yaml)
3. Refactor existing scripts to read from configuration
4. Add validation functions for configuration integrity

Phase 2: Function Modularity

1. Refactor calculate_carryover() to use configuration object
2. Add individual-level half-life heterogeneity option
3. Implement additional decay models (power law, two-compartment)
4. Create unit tests for each decay model

Phase 3: Empirical Integration

1. Build carryover estimation pipeline (Goal 2 deliverables)
2. Create import_carryover_estimate() function
3. Add validation for imported parameters
4. Document workflow for data-driven carryover specification

Phase 4: Sensitivity Framework

1. Implement run_carryover_sensitivity() wrapper
2. Create visualization functions for sensitivity results
3. Add parallel execution support for large grids
4. Generate sensitivity analysis vignette

12.3.4 Deliverables for Goal 3

1. Externalized carryover configuration system (YAML + R interface)
2. Refactored calculate_carryover() with modular design
3. Import pathway for empirically-estimated carryover parameters
4. Sensitivity analysis framework with visualization
5. Updated vignettes demonstrating new modularity features
6. Unit test coverage for all carryover functions

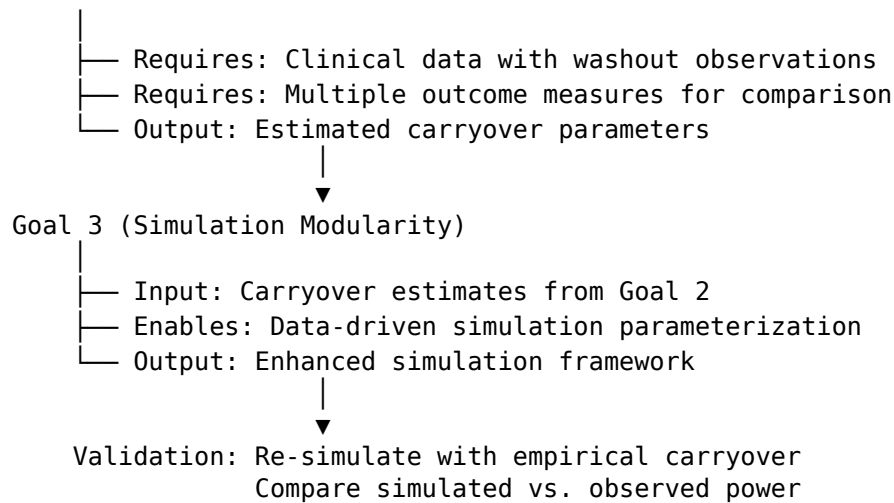
12.4 Integration Timeline and Dependencies

Goal 1 (Prazosin Efficacy)

- ├─ Requires: Clinical data import
- ├─ Requires: Carryover specification (from Goal 2 or assumed)
- └─ Output: Treatment effect estimates



Goal 2 (Carryover Characterization)



Dependency Notes:

- Goal 1 can proceed with assumed carryover values, then be re-run after Goal 2
- Goal 2 requires post-discontinuation data points to estimate decay
- Goal 3 benefits from Goal 2 but can proceed in parallel with placeholder values

12.5 Prazosin-Specific Considerations

12.5.1 Mechanism of Action Prazosin is an $\alpha 1$ -adrenergic receptor antagonist. In PTSD:

- Reduces noradrenergic hyperactivity during sleep
- Primary effect on trauma-related nightmares
- Secondary effects on hyperarousal symptoms
- Mechanism suggests rapid offset (receptor binding ~hours) but potentially prolonged clinical effect (neuroadaptation ~weeks)

12.5.2 Expected Carryover Profile Based on pharmacology and clinical experience:

Component	Expected Carryover	Rationale
Nightmare reduction	Short (days)	Direct receptor blockade
Sleep quality	Short-medium	Secondary to nightmare reduction
Daytime arousal	Medium (1-2 weeks)	Sustained neuroadaptation
Overall PTSD score	Medium	Composite of above

12.5.3 Biomarker Candidates For treatment moderation analysis, consider:

- Baseline PTSD severity (dose-response relationship)
- Nightmare frequency at baseline (target symptom)
- Autonomic reactivity measures (physiological target)
- Trauma type/chronicity (moderates treatment response)
- Prior medication history (sensitization effects)

12.6 Summary: Three-Goal Integration

Goal	Primary Question	Key Method	Output
1	Did prazosin work?	Mixed-effects model with carryover	Effect estimates, p-values
2	How long does effect persist?	Empirical carryover estimation	Half-life estimate, decay model
3	Can simulation use real parameters?	Modular configuration system	Enhanced simulation framework

The three goals form a coherent pipeline: clinical data analysis (Goal 1) informs carryover characterization (Goal 2), which parameterizes improved simulations (Goal 3), enabling validation of the analytical approach against known ground truth.

Appendix A: Covariance Matrix Construction

The framework implements a three-stage partitioned approach for constructing positive-definite covariance matrices:

1. **Stage 1:** Within-component temporal correlations (AR(1) structure)
2. **Stage 2:** Cross-component correlations (same-time and different-time)
3. **Stage 3:** Biomarker-baseline-response correlations

Eigenvalue validation ensures positive definiteness before simulation proceeds.

Appendix B: Carryover Decay Models

Three decay models are supported:

1. **Exponential** (default): $(1/2)^{(\text{scale} \times t / t_{\text{half}})}$
2. **Linear**: $\max(0, 1 - t / t_{\text{half}})$
3. **Weibull**: $\exp(-(t / t_{\text{half}})^{\text{shape}})$

The exponential model aligns with pharmacokinetic first-order elimination and is recommended for most applications.

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